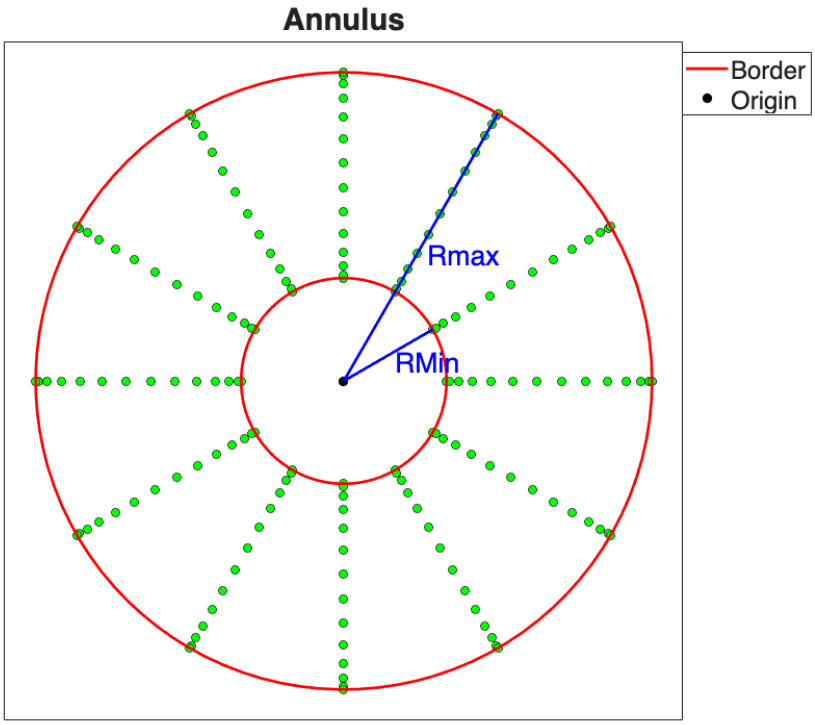
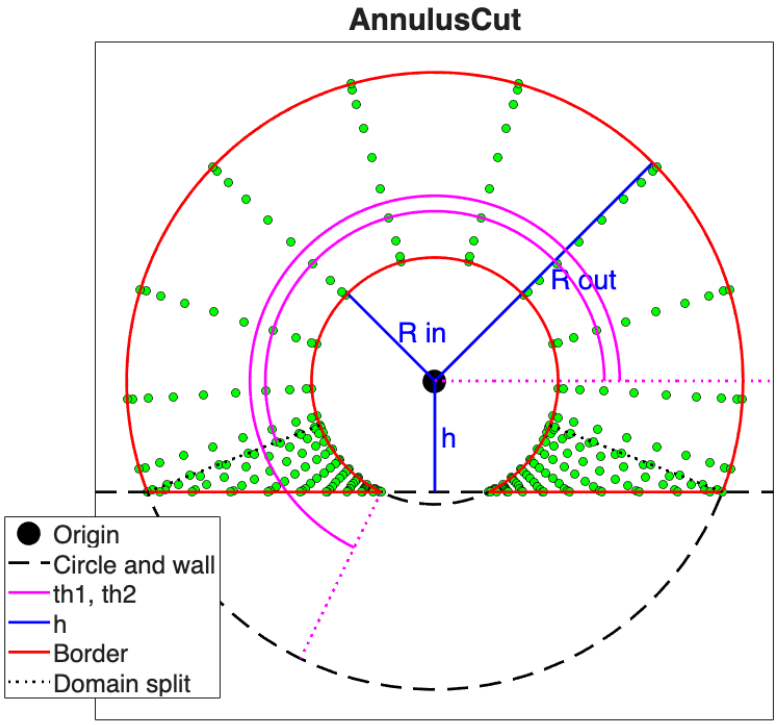


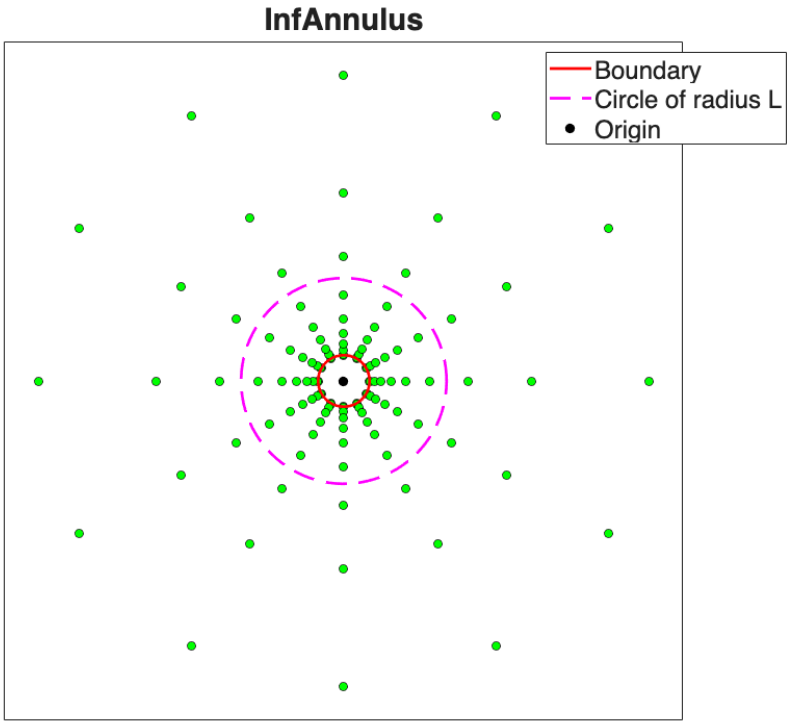
Annulus
An annulus with inner radius RMin and outer radius RMax, centred at Origin.
Note that this is defined in polar coordinates with y1 the radius (on [RMin,RMax]) and y2 the angle (on [0,2pi)).

Inputs:
N - [N1,N2] the number of points in the two coordinates
RMin (or R_in) - the inner radius of the annulus
RMax (or R_out) - the inner radius of the annulus
(Origin = [0,0]) - [O1,O2] the origin of the annulus



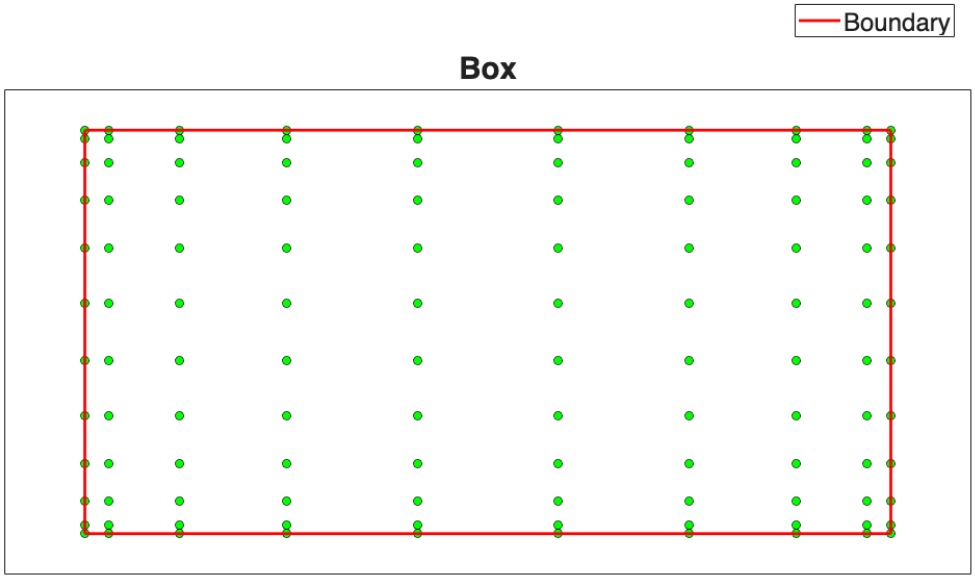

```
AnnulusCut
  An annulus with innner radius R_in and outer radius R_in cut off by a
  wall that is h below the Origin.  This is a ComposedShape with
  SubShape structure.

  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R_out - outer radius of the annulus
   R_in  - inner radius of the annulus
   h - perpendicular distance from the Origin to the wall
   (Origin = [0,0]) - origin of the annulus
```



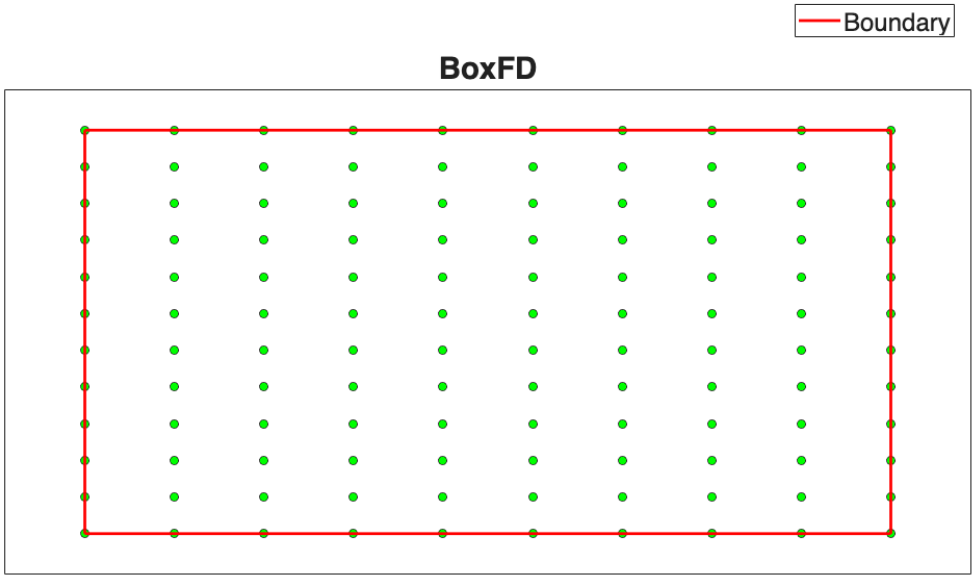

```
InfAnnulus
  An infinite annulus with inner radius RMin and outer radius inf, centred
  at Origin.
  Note that this is defined in polar coordinates with y1 the radius (on
  [RMin,inf]) and y2 the angle (on [0,2pi)).

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   RMin - the inner radius of the annulus
   L - a mapping parameter such that half of the radial points lie in
   [0,L]
   (Origin = [0,0]) - [O1,O2] the origin of the annulus
```



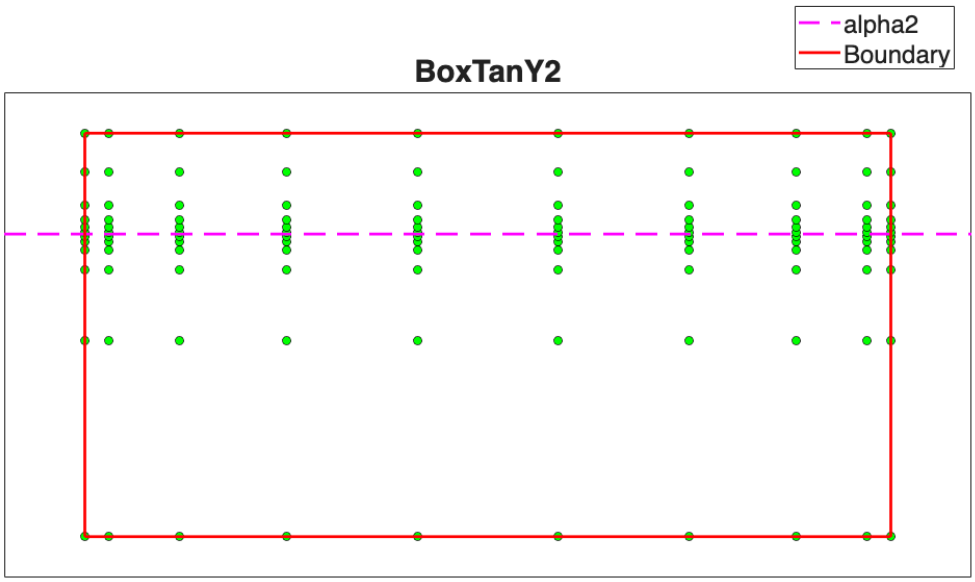

```
Box
  A box with corners defined by (y1Min,y2Min), (y1Max,y2Min),
  (y1Min,Y2Max), and (y1Max,y2Max).
  Alternatively, specify the lengths of the two sides, L1 and L2, and
  (optionally) the Origin, which is the bottom left of the box

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)       - left edge
   y1Max (or L1)     - right edge
   (y2Min = 0)       - bottom
   y2Max (or L2)     - top
   (Origin = [0,0]) - bottom left of the box when using L1, L2
```




```
BoxFD
  A box with corners defined by (y1Min,y2Min), (y1Max,y2Min),
  (y1Min,Y2Max), and (y1Max,y2Max).
  Alternatively, specify the lengths of the two sides, L1 and L2, and
  (optionally) the Origin, which is the bottom left of the box.
  NOTE: THIS USES FINITE DIFFERENCE

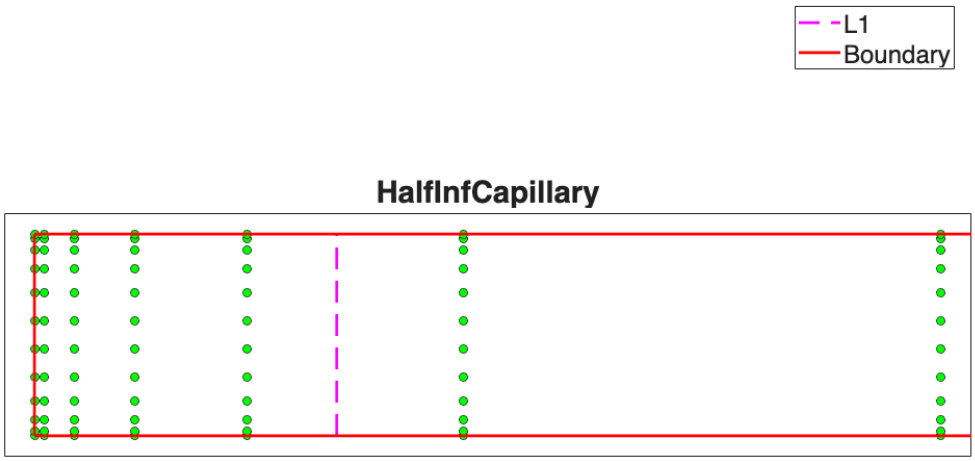
  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)      - left edge
   y1Max (or L1)    - right edge
   (y2Min = 0)      - bottom
   y2Max (or L2)    - top
   (Origin = [0,0]) - bottom left of the box when using L1, L2
```




```
BoxTanY2
  A box with corners defined by (y1Min,y2Min), (y1Max,y2Min),
  (y1Min,Y2Max), and (y1Max,y2Max).
  Alternatively, specify the lengths of the two sides, L1 and L2, and
  (optionally) the Origin, which is the bottom left of the box

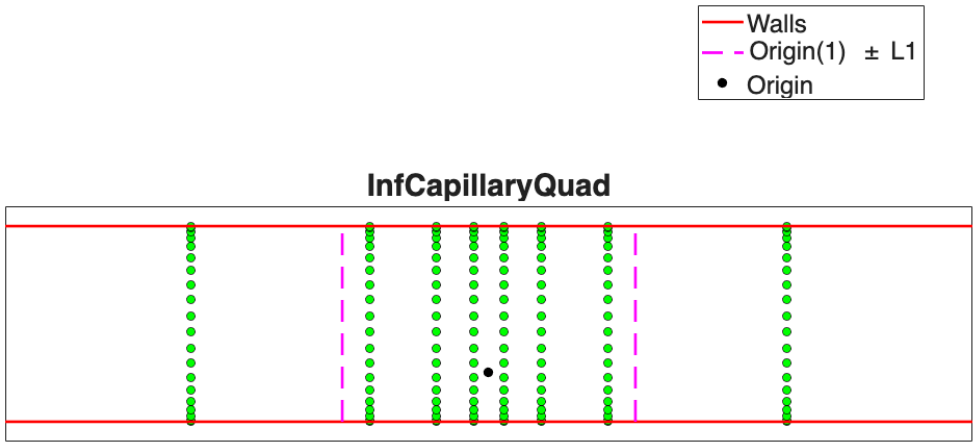

  This differs from BOX by using a tangent-based map in y2 to cluster
  the points around a given value - essentially an adapted grid.
  If alpha1>>1, points cluster around alpha2.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)       - left edge
   y1Max (or L1)     - right edge
   (y2Min = 0)       - bottom
   y2Max (or L2)     - top
   (Origin = [0,0]) - bottom left of the box when using L1, L2
   alpha1   - parameter for tan map
   alpha2   - parameter for tan map
```



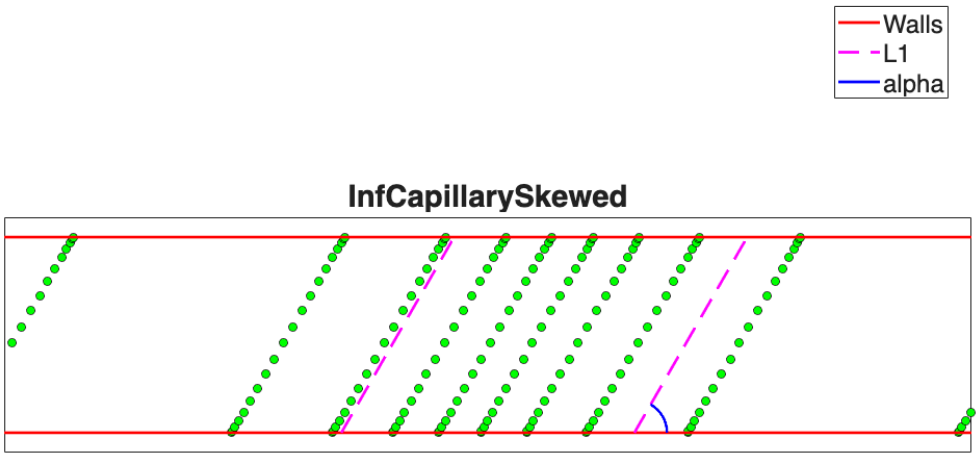

```
HalfInfCapillary
  A capillary that is half infinite in the first coordinate and finite
  in the second, with bounds given by y2Min and y2Max.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - mapping parameter for first coordinate.  Half the points lie in
        [0,L1]
   y2Min - bottom
   y2Max - top
```



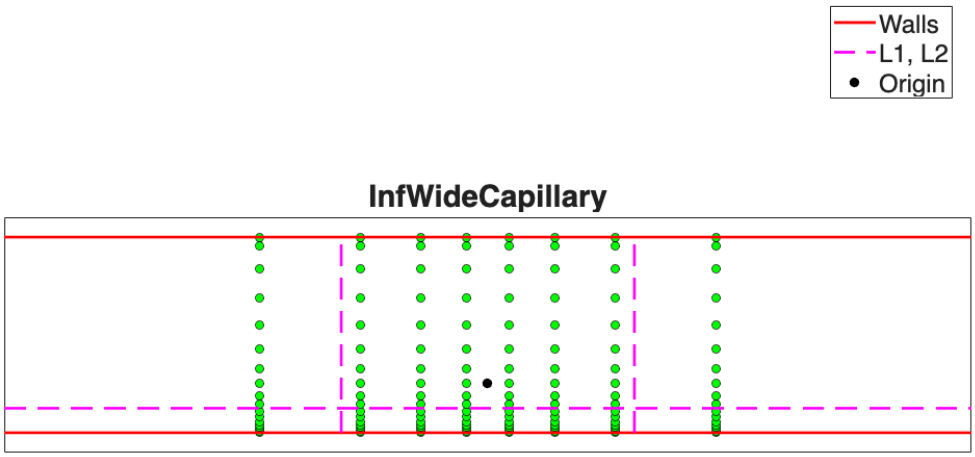

```
InfCapillaryQuad
  An infinite (in the first coordinate) capillary, which is finite in
  the second coordinate between y2Min and y2Max.
  Note this only differs from InfCapillary through the use of a
  quadratic map (rather than square root) in the first coordinate.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - mapping parameter for the first coordinate
   y2Min - bottom
   y2Max - top
   y10 - origin in first coordinate
   (Conv = false) - true/false to set up for FMT-like convolution
```

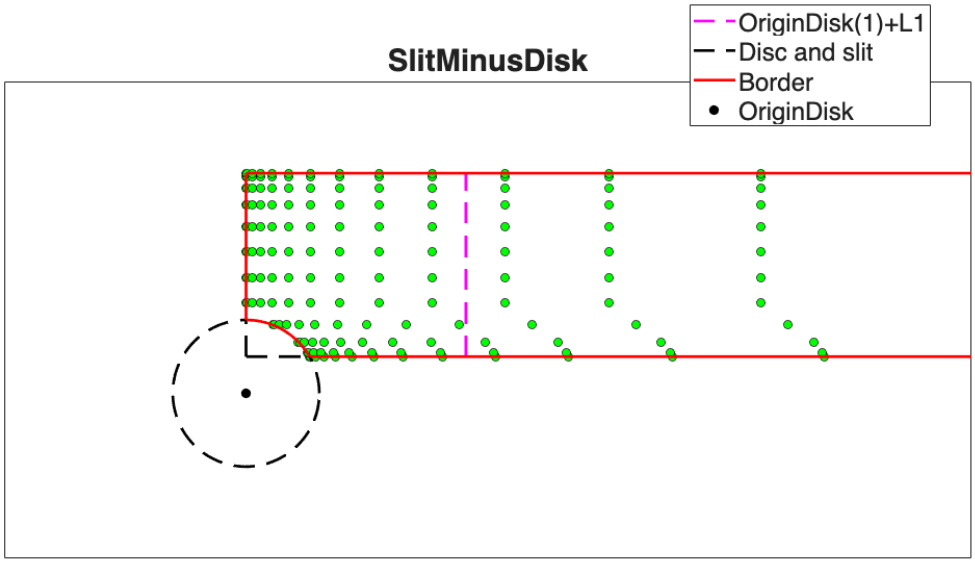



```
InfCapillarySkewed
  An infinite (in the first coordinate) capillary, which is finite in
  the second coordinate between y2Min and y2Max.  The coordinate grid
  is skewed at an angle alpha.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - mapping parameter for the first coordinate
   y2Min - bottom
   y2Max - top
   y10 - origin in first coordinate
   (Conv = false) - true/false to set up for FMT-like convolution
   (alpha = pi/2) - skew of grid
```

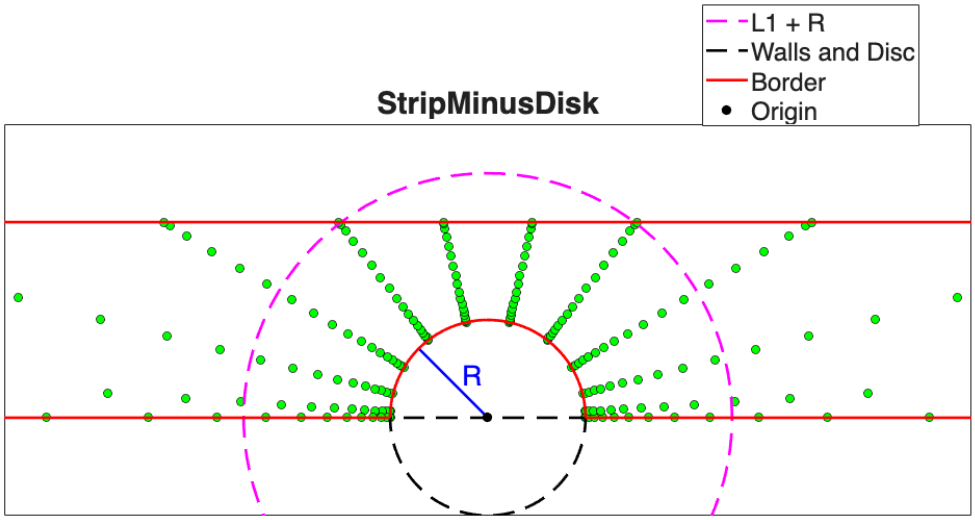



```
InfWideCapillary
  File: InfWideCapillary.m
  Created Date: 16 Oct 2025
  Author: Jake Skelton
  Date Modified: Thu Oct 16 2025
  Modification of InfCapillaryGeneral for a capillary wide in the finite
direction
```



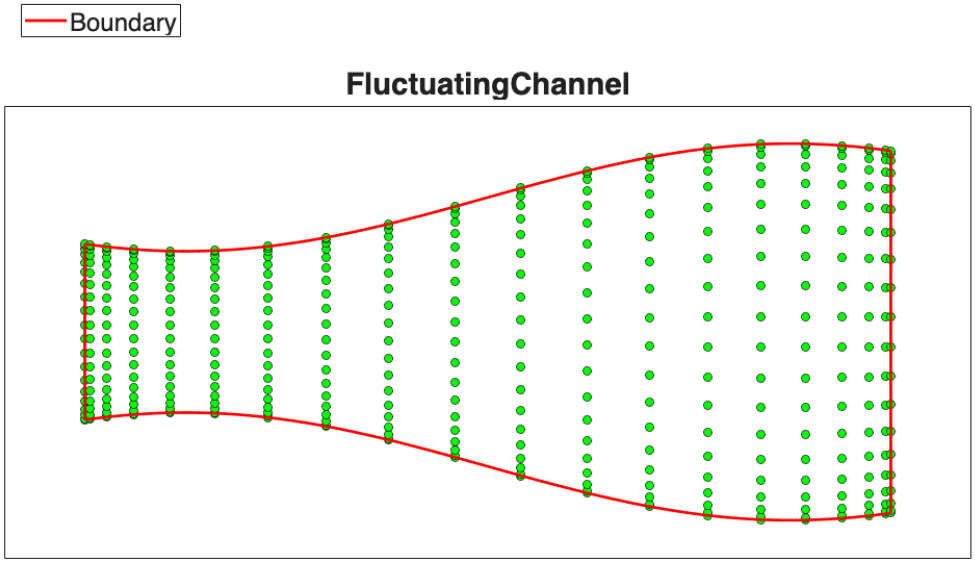

```
SlitMinusDisk
  Part of a half-infinite slit (either 'Left' or 'Right') cut off at
  one end by a disc of radius R centred at OriginDisk.  The finite end
  of the strip is at OriginDisk(1).

  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R - radius of the disc to be removed
   L1 - mapping parameter for radial coordinate; half the points lie in
        [R,R+L1]
   y2Min, y2Max - positions of the upper and lower walls
   (LeftRight = 'Right') - 'Left' or 'Right' for direction of slit
   (OriginDisk = [0,0]) - origin of the disc
```



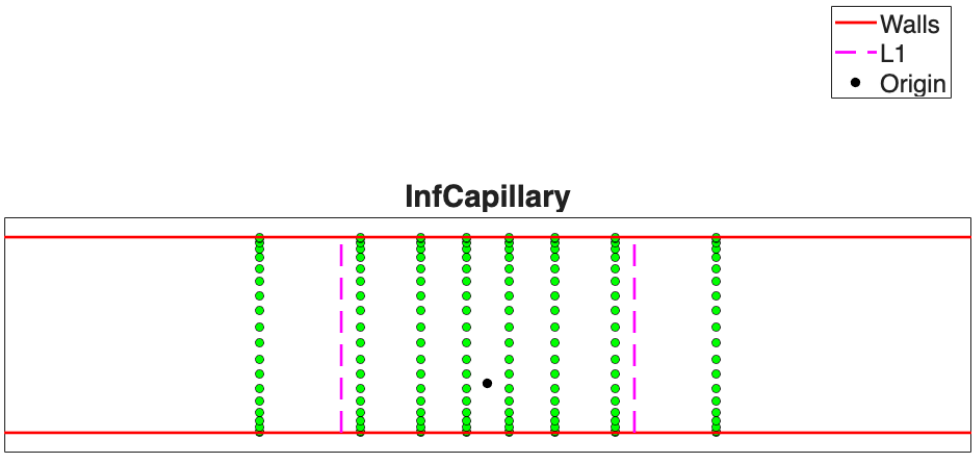

```
StripMinusDisk
  An infinite, horizontal strip with a disc of radius R centred at
  Origin removed.  The other edge of the strip is at y2Wall.  The half
  disc must fit in the strip, otherwise use HalfStripMinusDisk.

  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R - radius of the disc to be removed
   L1 - mapping parameter for radial coordinate; half the points lie in
        [R,R+L1]
   y2Wall - position of lower wall
   (TopBottom = 'Bottom') - 'Top' or 'Bottom' for whether strip is at
         the Top or Bottom of the disc.
   (Origin = [0,0]) - origin of the disc
```



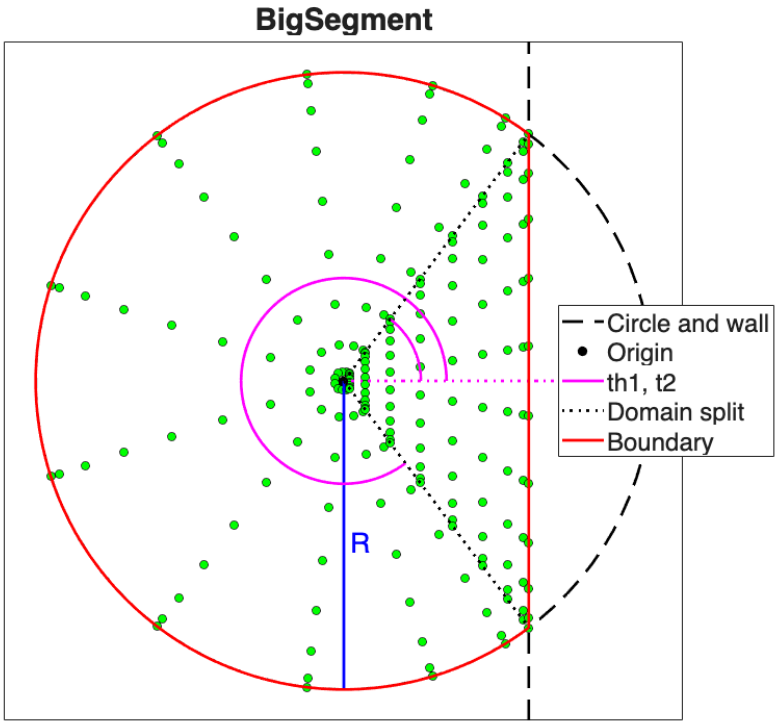
FluctuatingChannel
A finite channel, of length L1, where the top and bottom are defined by functions, ft and fb.

Inputs:
N - [N1,N2] the number of points in the two coordinates
ft - function name for top; input is (x,params), output is y
fb - function name for bottom; input is (x,params), output is y
ftparams - parameters for ft
fbparams - parameters for fb




```
InfCapillary
  An infinite (in the first coordinate) capillary, which is finite in
  the second coordinate between y2Min and y2Max.

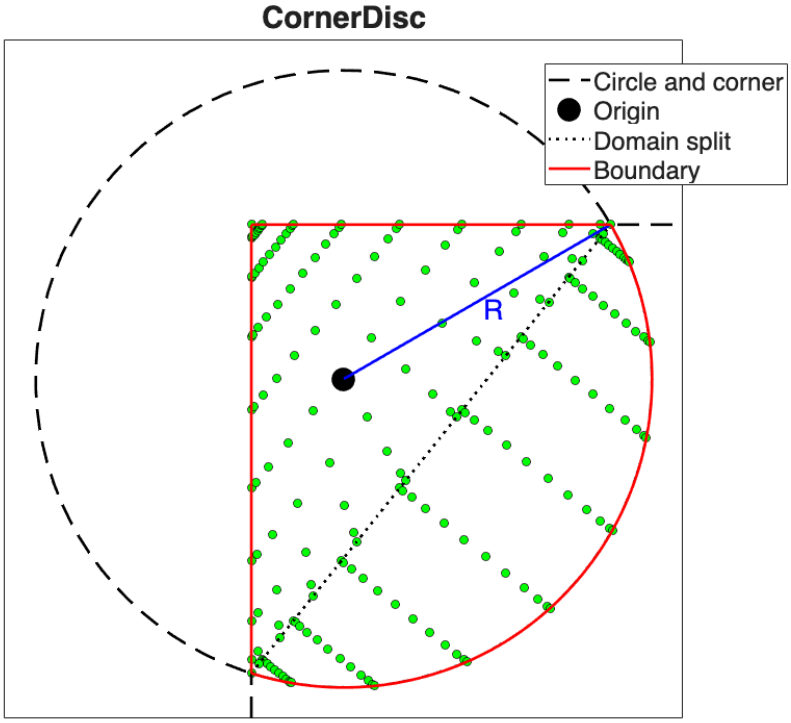
  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - mapping parameter for the first coordinate
   y2Min - bottom
   y2Max - top
   y10 - origin in first coordinate
   (Conv = false) - true/false to set up for FMT-like convolution
```




```
BigSegment
  A 'big' segment of a disc of radius R, created by a
  horizontal/vertical chord at perpendicular position Wall_Y.
  Note that this is the complement of Segement.

  This is formed from a triangle and wedge (the part of a disc between
  two angles).

  Inputs:
    NT - [NT1,NT2] number of points for the triangle in the two coordinates
    NW - [NW1,NW2] number of points for the wedge in the two
    coordinates (radial and angular)
    R - radius of the disc
    Wall_VertHir - 'horizontal' or 'vertical' orientation of the wall
    Wall_Y - coordiate of the wall in the perpendicular direction
    Wall - one of 'N', 'S', 'E', 'W' denoting the compass direction of
    the wall from the origin.  Only used when we have exactly half a
    disc
    (Origin = [0,0]) - origin of the disc
    (sphere = false) - true implements 'spherical integration'
```

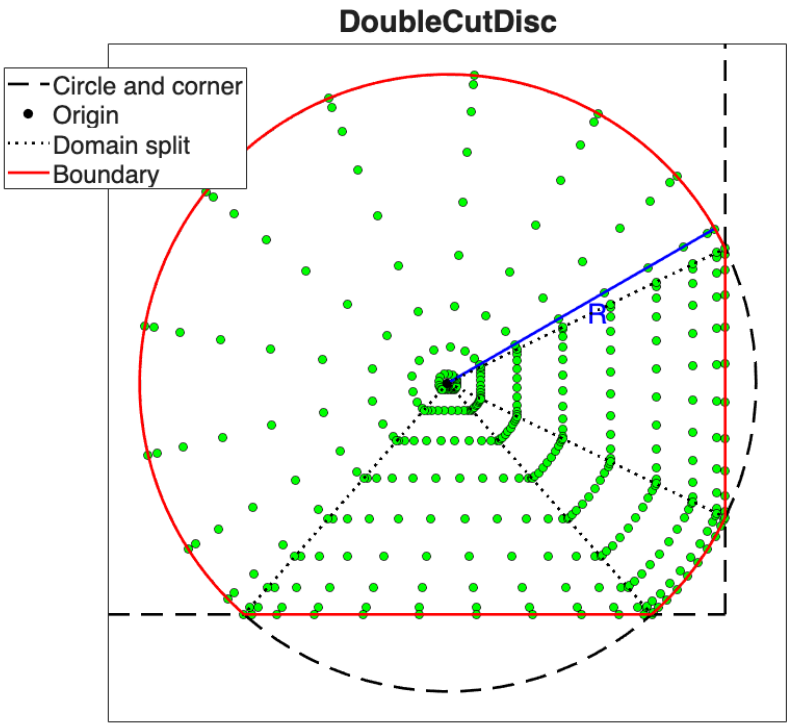



```
CornerDisc
  Part a disc of radius R, centred at Origin, created by it being cut
  off by the corner of an abritrarily large square, with the corner at
  CornerPos.

  Note that the corner must lie inside the disc.  Otherwise try
  DoubleCutDisc.

  This is formed from a triangle and segment.

  Inputs:
    NT - [NT1,NT2] number of points for the triangle in the two coordinates
    NS - [NS1,NS2] number of points for the segment in the two
    coordinates
    R - radius of the disc
    (Corner = 'SW') - one of 'NE', 'SE', 'SW', 'NW' denoting the compass
direction of
    the corner from the centre of the square
    (CornerPos = [0,0]) - [CP1,CP2] the position of the corner
    (Origin = [0,0]) - origin of the disc
```

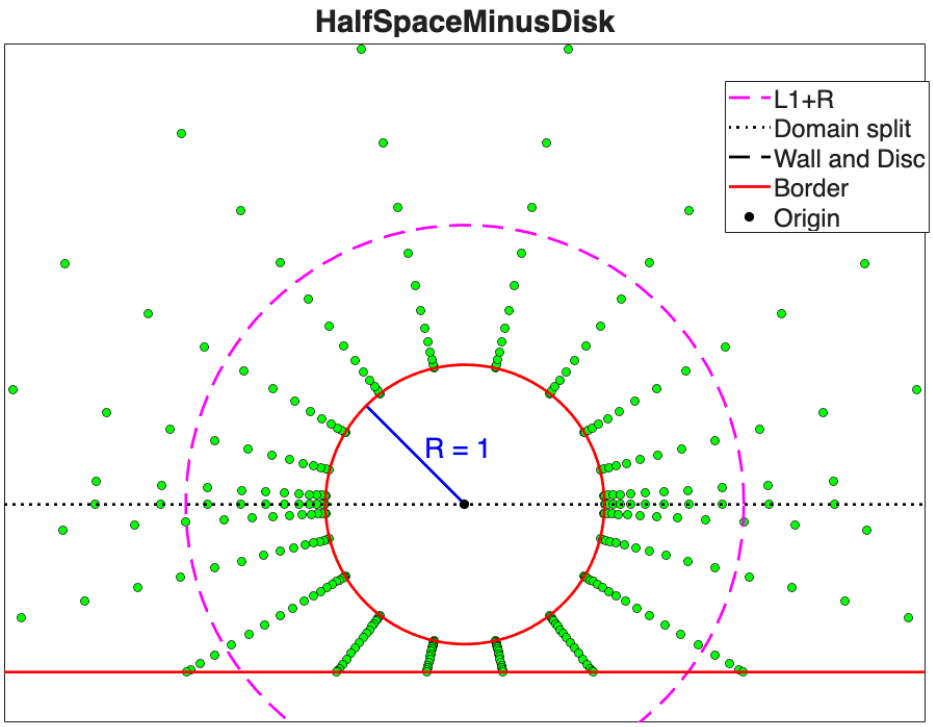



```
DoubleCutDisc
  Part a disc of radius R, centred at Origin, created by it being cut
  off by the corner of an abritrarily large square, with the corner at
  CornerPos.

  Note that the corner must lie outside the disc.  Otherwise try
  CornerDisc.

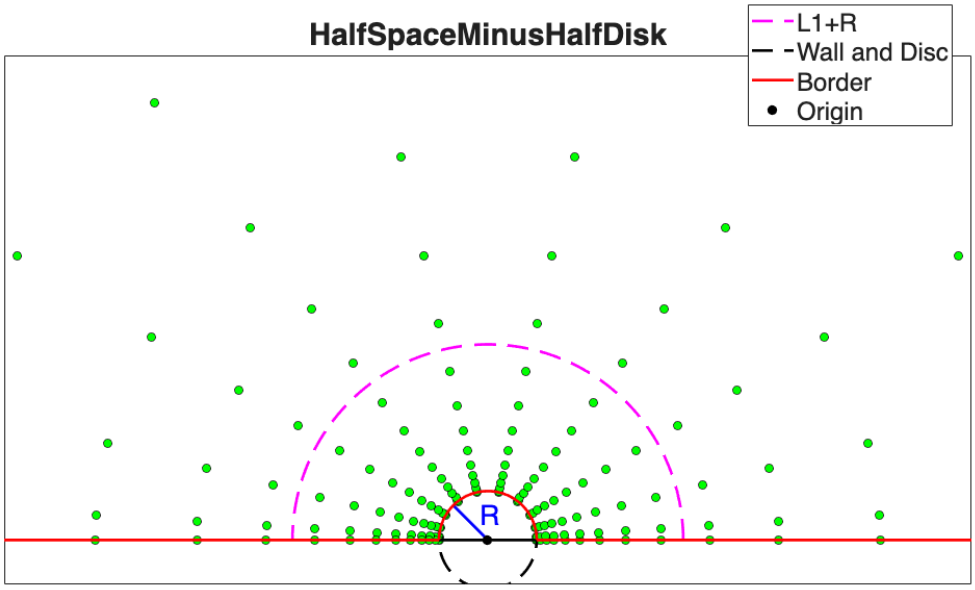
  This is formed from two triangles and two wedges

  Inputs:
    NT - [NT1,NT2] number of points for the triangles in the two coordinates
    NW - [NS1,NS2] number of points for the wedges in the two coordinates
    R - radius of the disc
    (Corner = 'SW') - one of 'NE', 'SE', 'SW', 'NW' denoting the compass
direction of
    the corner from the centre of the square
    (CornerPos = [0,0]) - [CP1,CP2] the position of the corner
    (Origin = [0,0]) - origin of the disc
```



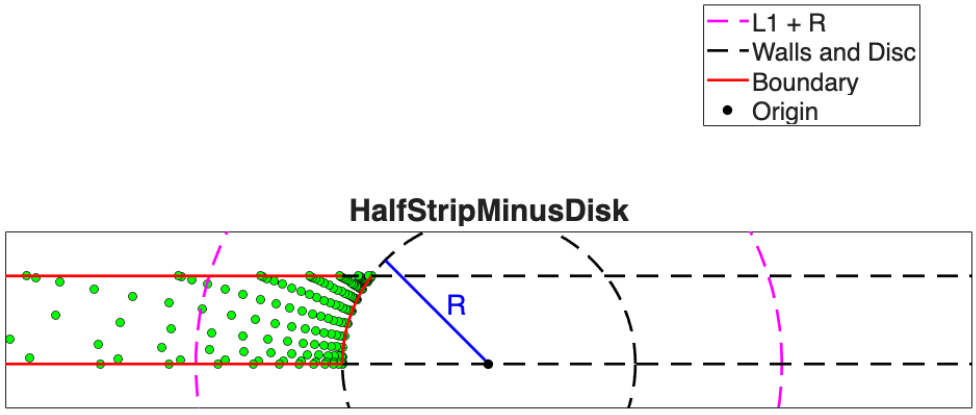

```
HalfSpaceMinusDisk
  An upper half space without a disc of radius R at Origin.
  This is a ComposedShape.
  Note that the wall, given by y2Wall, must be below the Origin of the
  disc.

  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R - radius of the disc to be removed
   y2Wall - location of the wall
   L1 - mapping parameter for radial coordinate; half the points lie in
        [R,R+L1]
   (Origin = [0,0]) - origin of the disc
```



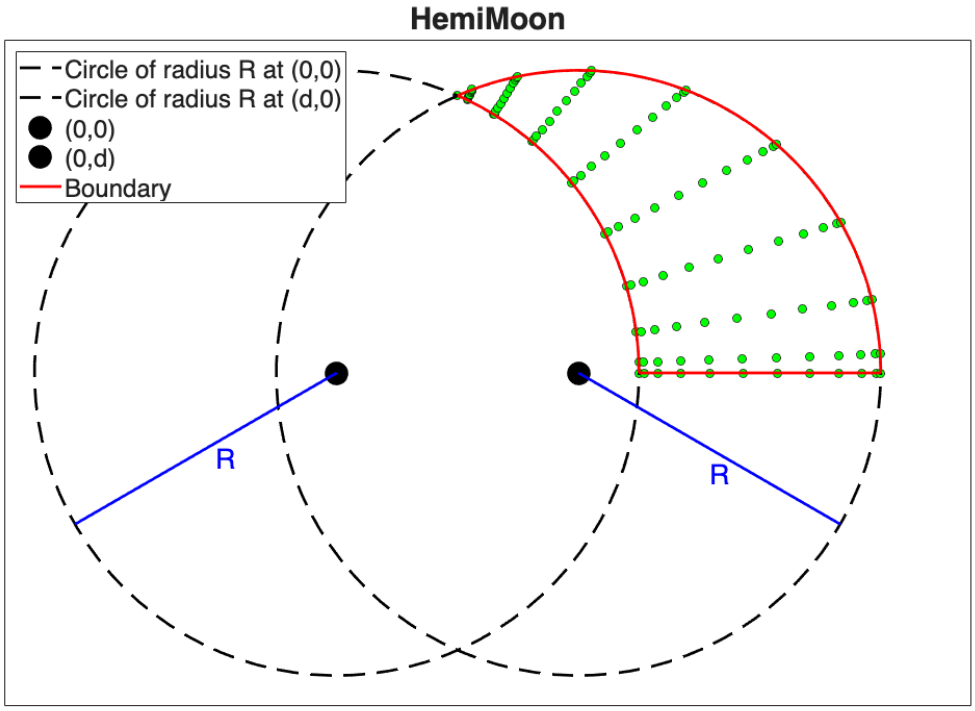

```
HalfSpaceMinusHalfDisk
  An half space from Origin(2) in the second coordinate with half a
  disc of radius R and centred at Origin removed.  Which half space is
  determined by TopBottom.  This is in polar coordinates.
  Can also specify Rmax, which results in half an annulus.

  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R - radius of the disc to be removed
   (Rmax = inf) - outer radius
   L1 - mapping parameter for radial coordinate; half the points lie in
        [R,R+L1]
   (Origin = [0,0]) - origin of the disc
```

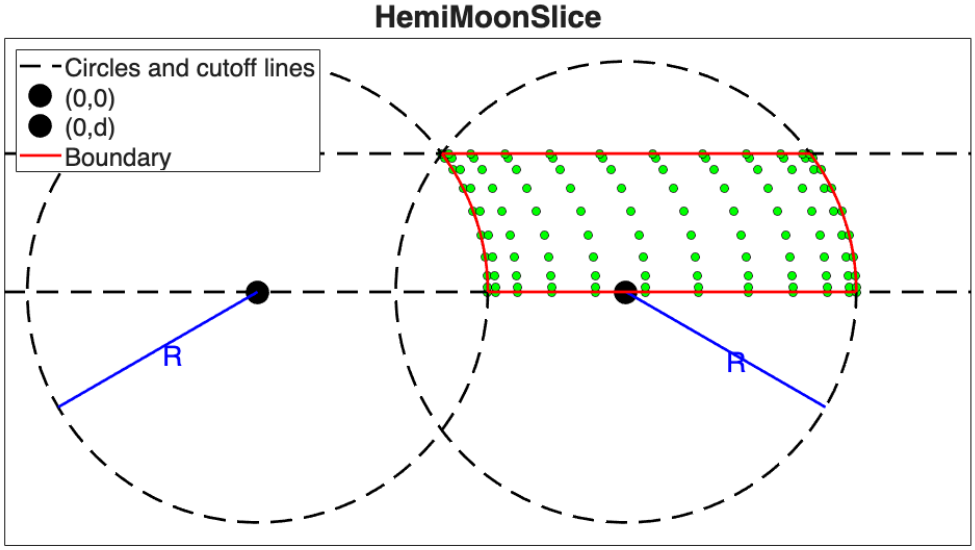



```
HalfStripMinusDisk
  Part of a half-infinite strip (either 'Left' or 'Right') cut off by a
  disc of radius R centred at Origin on either the 'Top' or 'Bottom' edge.
  The other edge of the strip is at y2Wall.

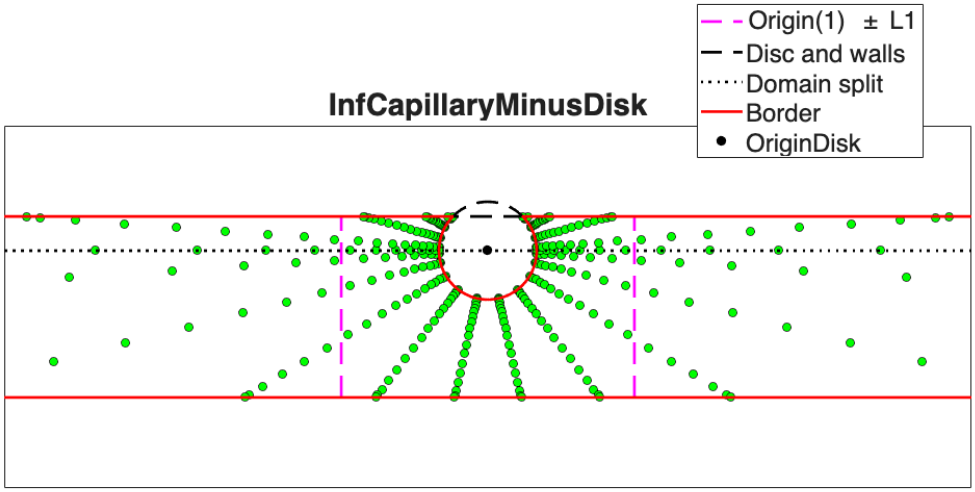
  Inputs:
   N - [N1,N2] number of points in the two coordinates
   R - radius of the disc to be removed
   L1 - mapping parameter for radial coordinate; half the points lie in
        [R,R+L1]
   y2Wall - position of lower wall
   LeftRight - 'Left' or 'Right' for direction of strip
   TopBottom - 'Top' or 'Bottom' for whether strip is at
         the Top or Bottom of the disc.
   (Origin = [0,0]) - origin of the disc
```


HemiMoon
Circle of radius R at (0,0)
Circle of radius R at (d,0)
(0,0)
(0,d)
Boundary
R
R

```
HemiMoon
 {
```

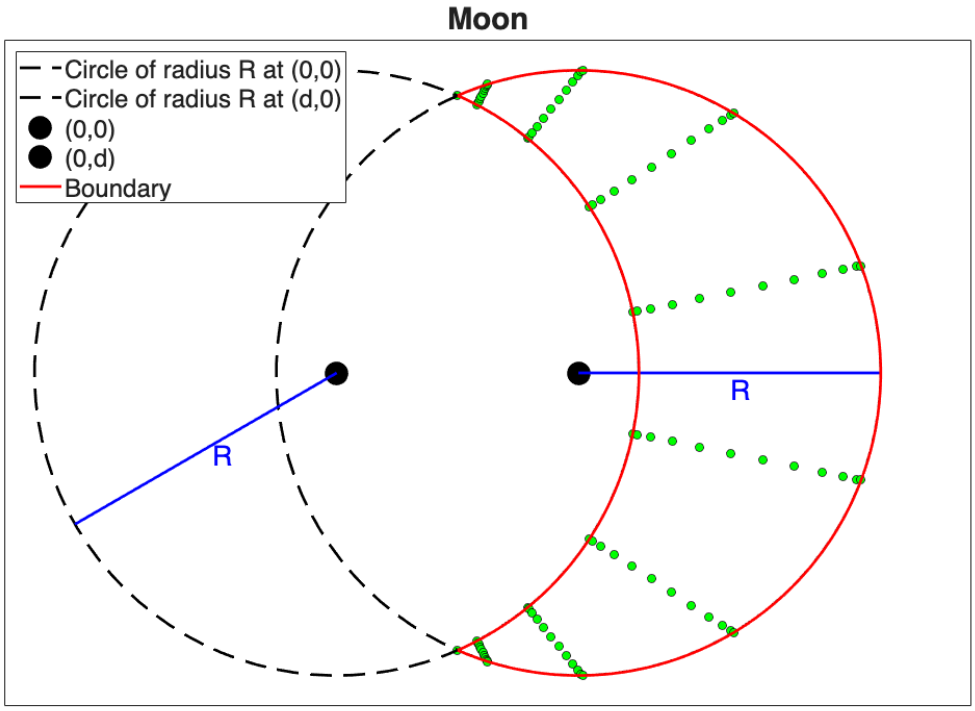



```
HemiMoonSlice
 {
```



InfCapillaryMinusDisk
An infinite (in the first coordinate) capillary with walls at y1Min and y2Max, with a disc centred at Origin removed. Note that the Origin must lie inside the capillary, and the disc must not completely disect the capillary.

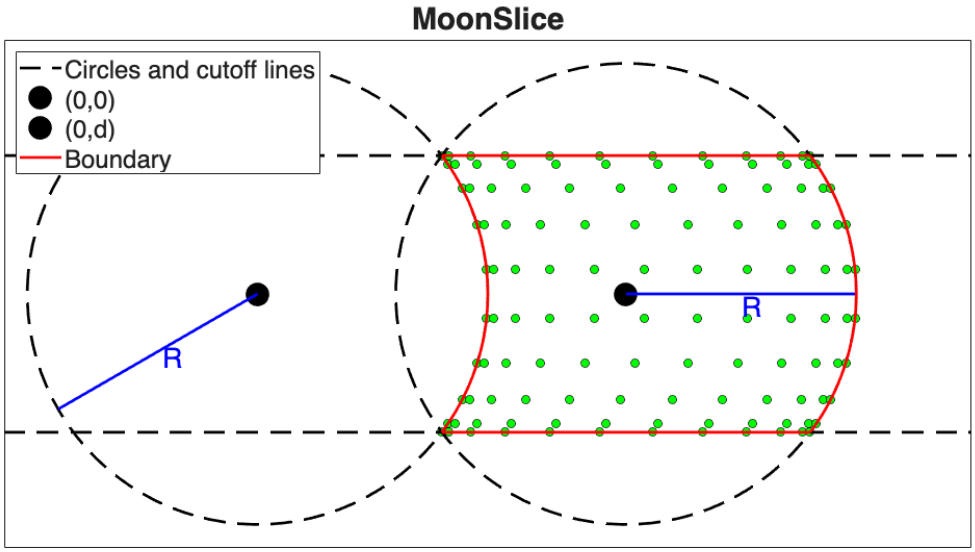
Inputs:
N - [N1,N2] number of points in the two coordinates
R - radius of the disc to be removed
L1 - mapping parameter for the first coordinate
y2Min, y2Max - positions of the upper and lower walls
(Origin = [0,0]) - origin of the disc



Moon

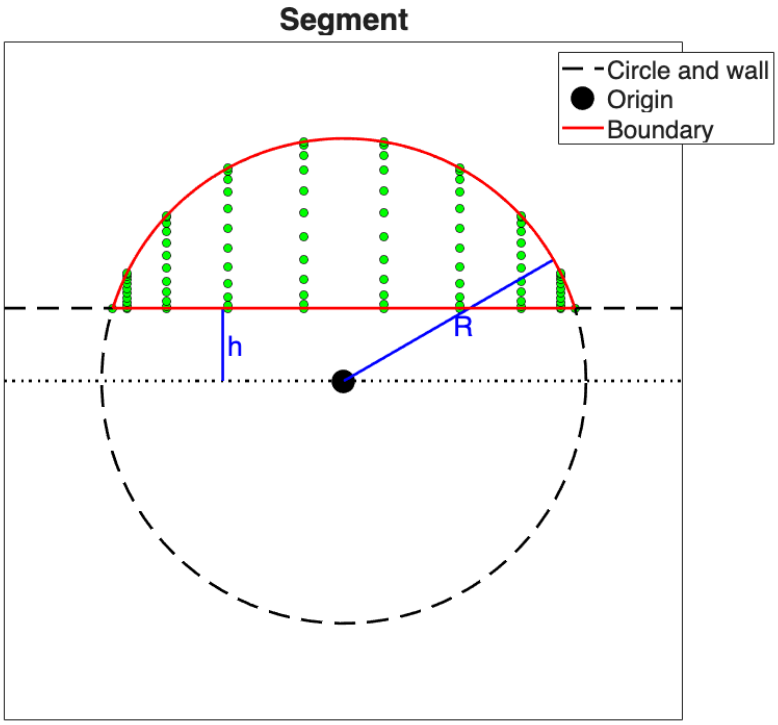
Part of a disc of radius R with origin (0,d) once the intersection with a disc of radius R with origin (0,0) is removed.
Note: must have d < R.
Note: Origin is hard coded to 0.

Inputs:
N - [N1,N2] number of points in the two coordinates
R - radii of the two discs
d - x-coordinate of the second disc



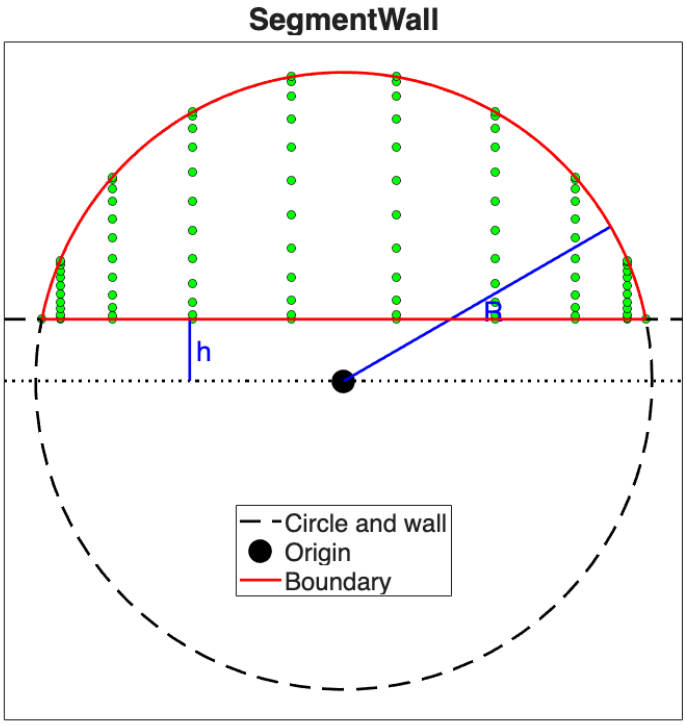
MoonSlice
Part of a disc of radius R with origin (0,d) once the intersection with a disc of radius R with origin (0,0) is remove and chopped off along the horizontal lines through the intersection points.
Note: Origin is hard coded to 0.

Inputs:
N - [N1,N2] number of points in the two coordinates
R - radii of the two discs
d - x-coordinate of the second disc



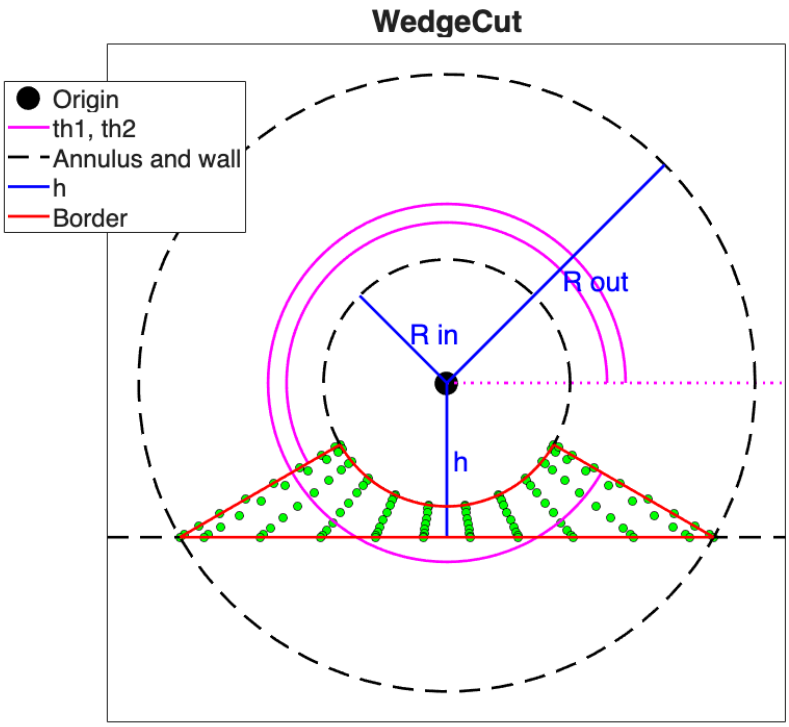

```
Segment
  A segment of a disc of radius R, created by a
  horizontal chord at (positive) distance h above the origin.
  To be used when h < R; otherwise use BigSegment.

  Inputs:
    N - [N1,N2] number of points in the two coordinates
    R - radius of the disc
    h - height of chord above the origin
    (Origin = [0,0]) - origin of the disc
    (sphere = false) - true implements 'spherical integration'
```



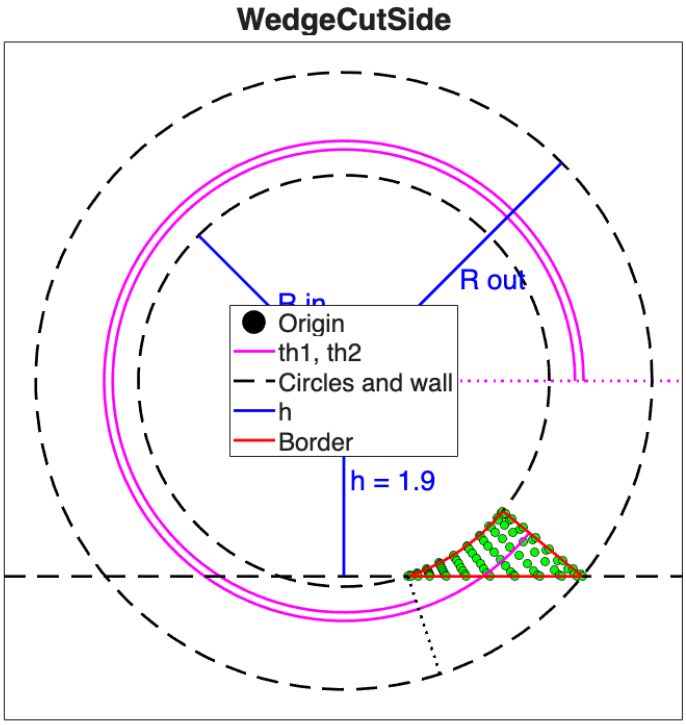

```
SegmentWall
  A segment of a disc of radius R, created by a
  horizontal or vertical chord (determined by Wall_VertHor),
  Wall_Y from the Origin.

  Inputs:
    N - [N1,N2] number of points in the two coordinates
    R - radius of the disc
    Wall_VertHor - 'vertical' or 'horizontal' orientation of wall
    Wall_Y - wall coordinate *not distance from wall*
    (Origin = [0,0]) - origin of the disc
    (sphere = false) - true implements 'spherical integration'
    (alpha) - fixed angle of rotation from the horizontal
```



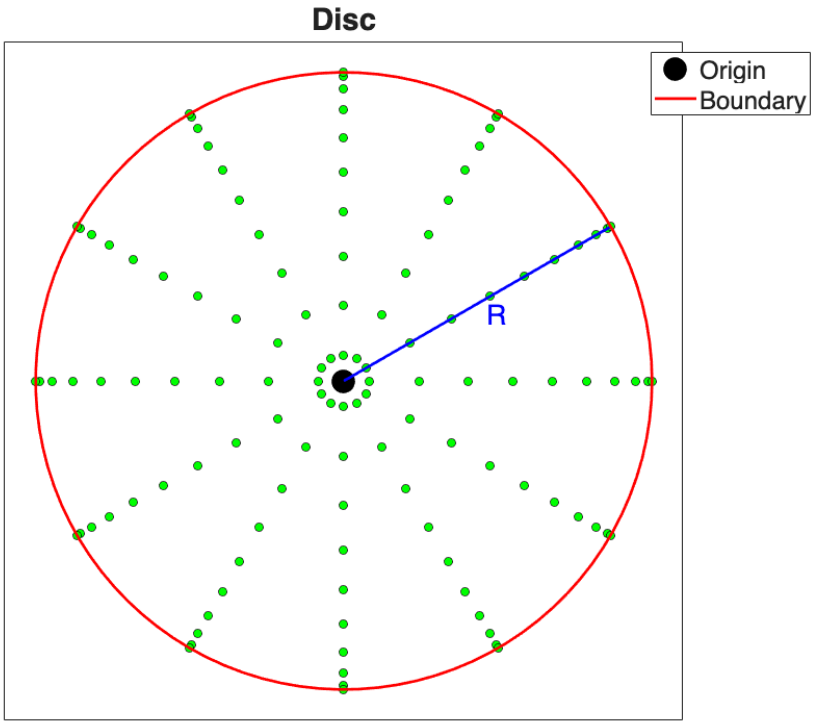
WedgeCut
A wedge of radius R_out cut off by a horizontal wall a distance h below the Origin. The angular range is such that the bottom corners of the WedgeCut are where the circle of radius R_out centred at Origin meets the wall.

Inputs:
N - [N1,N2] number of points in the two coordinates
R_out (or R) - outer radius of the wedge
(R_in = 0) - inner radius of the wedge
h - perpendicular distance from the Origin to the wall
(Origin = [0,0]) - origin of the wedge



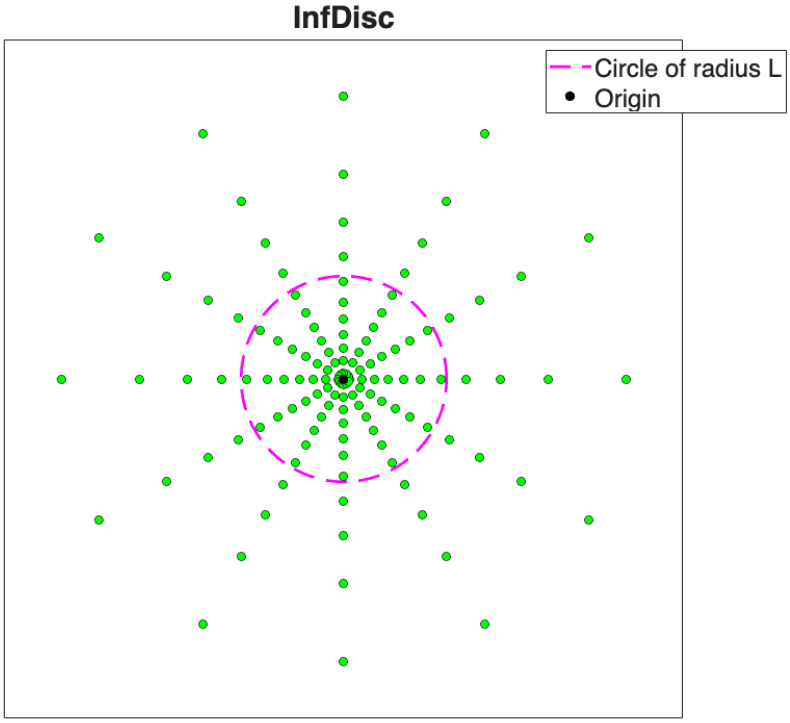
WedgeCutSide
A wedge of radius R_out cut off by a horizontal wall a distance h below the Origin. The angular range is such that the bottom corners of the WedgeCut are where the circle of radii R_in and R_out centred at Origin meet the wall.

Inputs:
N - [N1,N2] number of points in the two coordinates
R_out (or R) - outer radius of the wedge
(R_in = 0) - inner radius of the wedge
h - perpendicular distance from the Origin to the wall
(Origin = [0,0]) - origin of the wedge



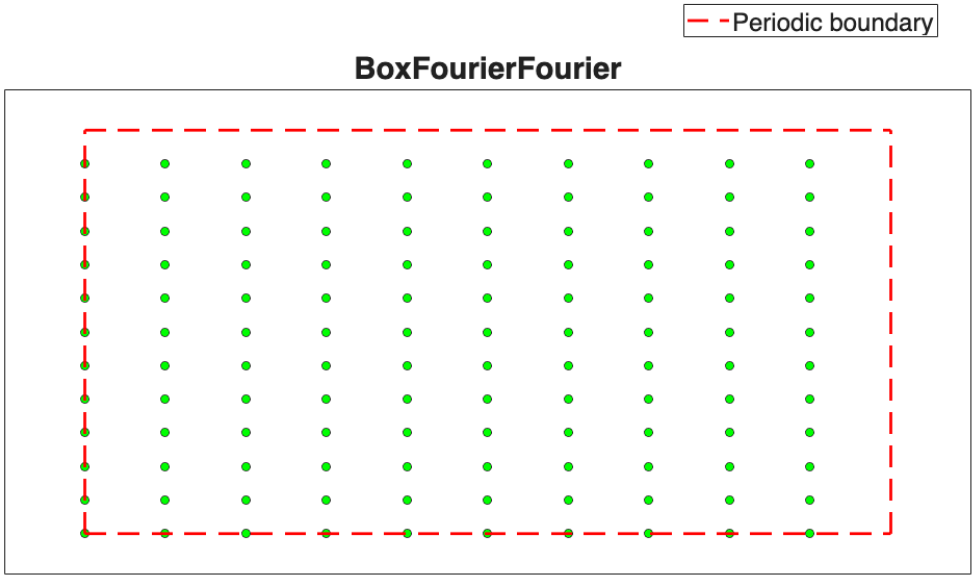
Disc
A disc of radius R, centred at Origin.
Note that this is defined in polar coordinates with y1 the radius (on (0,R]) and y2 the angle (on [0,2pi)).

Inputs:
N - [N1,N2] the number of points in the two coordinates
R - the radius of the disc
(Origin = [0,0]) - [O1,O2] the origin of the disc
(spherical = false) - whether to do spherically-weighted integration



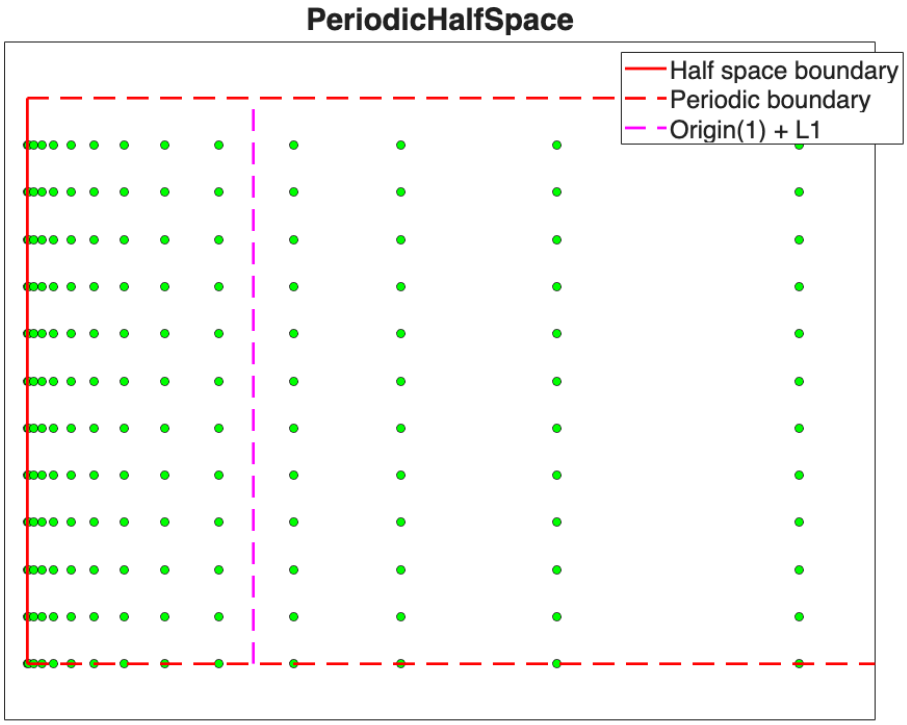

```
InfDisc
  A disc of infinite radius, centred at Origin.  I.e., the plane in
  polar coordinates.
  Note that this is defined in polar coordinates with y1 the radius (on
  (0,inf]) and y2 the angle (on [0,2pi)).

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L - a mapping parameter such that half of the radial points lie in
   [0,L]
   (Origin = [0,0]) - [O1,O2] the origin of the disc
```



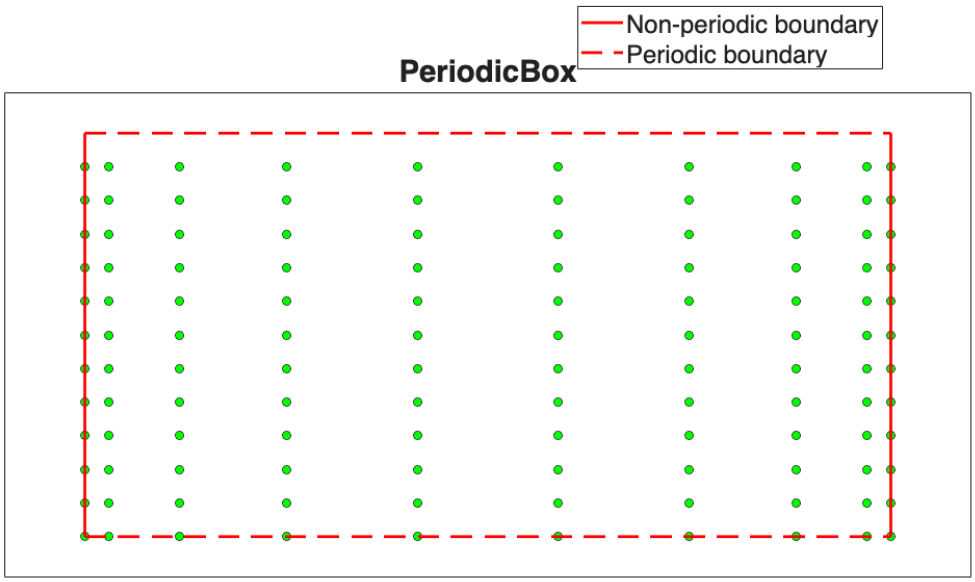

```
BoxFourierFourier
  A periodic box with corners defined by
  (y1Min,y2Min), (y1Max,y2Min), (y1Min,Y2Max), and (y1Max,y2Max).
  Alternatively, specify the lengths of the two sides, L1 and L2, and
  (optionally) the Origin, which is the bottom left of the box.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)      - left edge
   y1Max (or L1)    - right edge
   (y2Min = 0)      - bottom
   y2Max (or L2)    - top
   (Origin = [0,0]) - bottom left of the box when using L1, L2
```



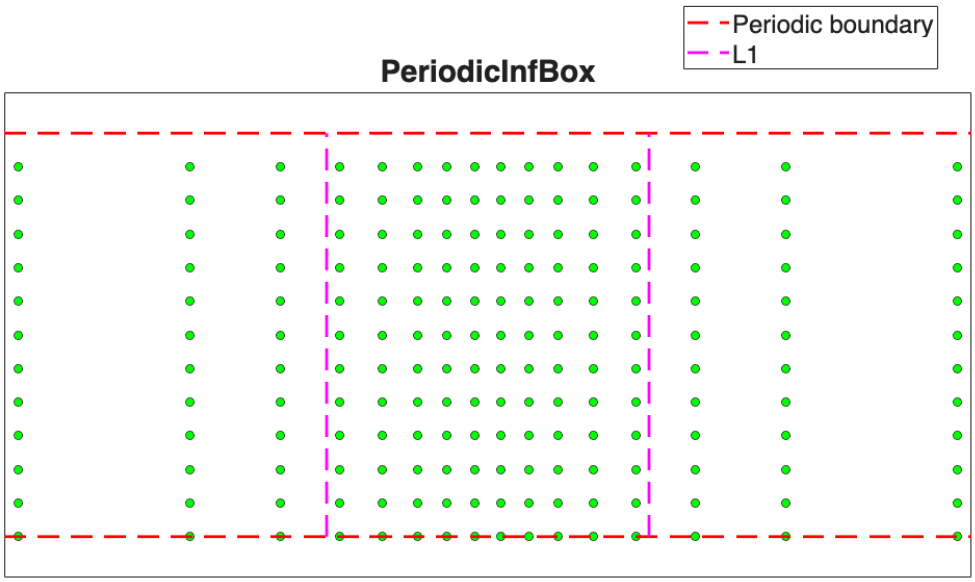

```
PeriodicHalfSpace
  A half space which is periodic in the second cooodinate, with
  length determined by y2Min and y2Max.
  Alternatively, specify the length of the second side, L2.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)   - left edge
   (y2Min = 0)   - bottom
   y2Max (or L2) - top
   L1            - mappring parameter for the first coordinate
```



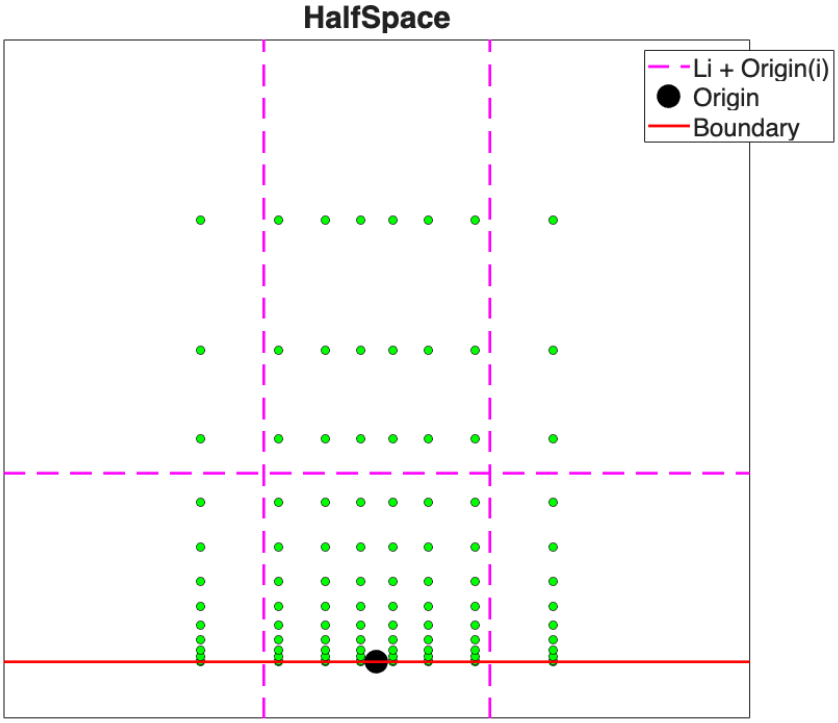

```
PeriodicBox
  A (half) periodic box with corners defined by
  (y1Min,y2Min), (y1Max,y2Min), (y1Min,Y2Max), and (y1Max,y2Max).
  Alternatively, specify the lengths of the two sides, L1 and L2, and
  (optionally) the Origin, which is the bottom left of the box
  The box is periodic in the second coordinate.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y1Min = 0)      - left edge
   y1Max (or L1)    - right edge
   (y2Min = 0)      - bottom
   y2Max (or L2)    - top
   (Origin = [0,0]) - bottom left of the box when using L1, L2
```



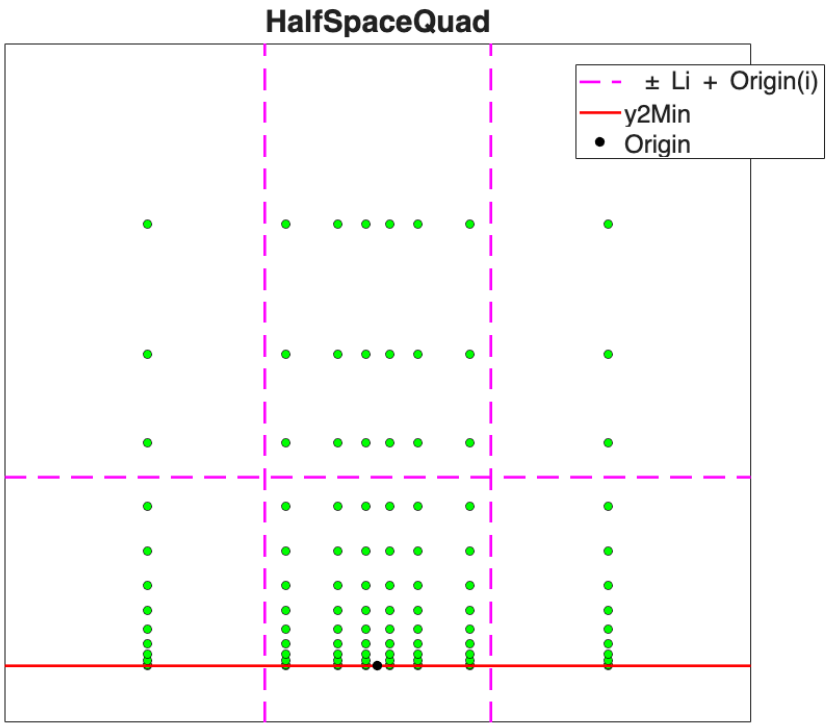

```
PeriodicInfBox
  A (half) periodic box which is infinite in the first coordinate.
  The box is periodic in the second coordinate, with length determined by
  y2Min and y2Max.
  Alternatively, specify the length of the second side, L2.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   (y2Min = 0)   - bottom
   y2Max (or L2) - top
   L1            - mapping parameter for the first coordinate
```



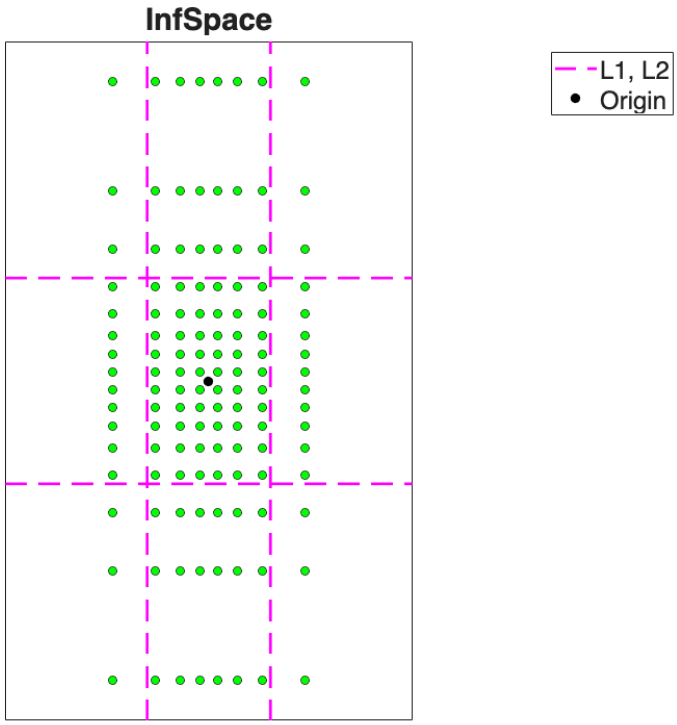

```
HalfSpace
  The upper half space ([-inf,inf] x [y2Min,inf] in Cartesian coordinates.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - a mapping parameter such that half of the y1 points lie in
   [-L1 + y10, L1 + y10]
   L2 - a mapping parameter such that half of the y2 points lie in
   [y2Min, y2Min + L2]
   (y10 = 0) - the 'origin' in y1
```



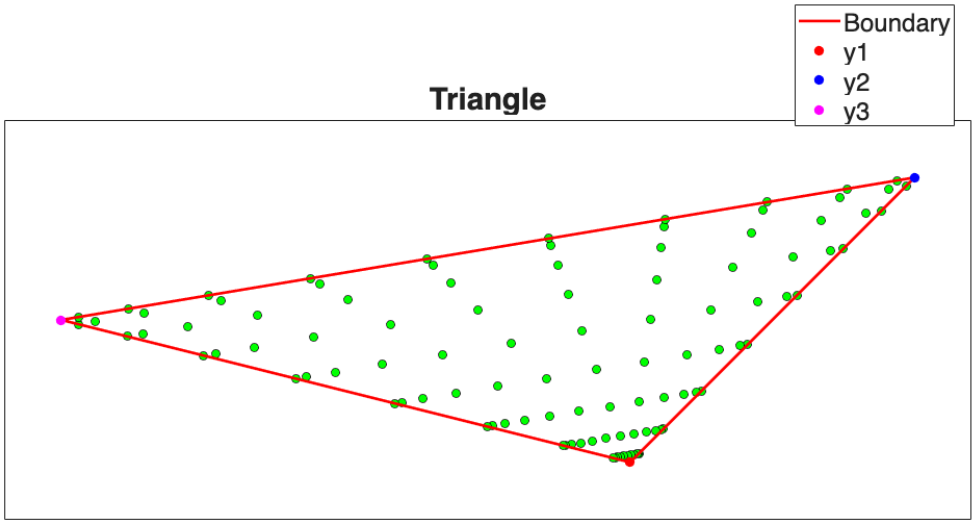

```
HalfSpaceQuad
  The upper half space ([-inf,inf] x [y2Min,inf] in Cartesian coordinates.
  Note that this is the same as HalfSpace except we use a quadratic map
  in the first coordinate, rather than a quotient map.
  Use HalfSpace unless you have a good reason not to.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - a mapping parameter such that half of the y1 points lie in
   [-L1 + y10, L1 + y10]
   L2 - a mapping parameter such that half of the y2 points lie in
   [y2Min, y2Min + L2]
   (y10 = 0) - the 'origin' in y1
```



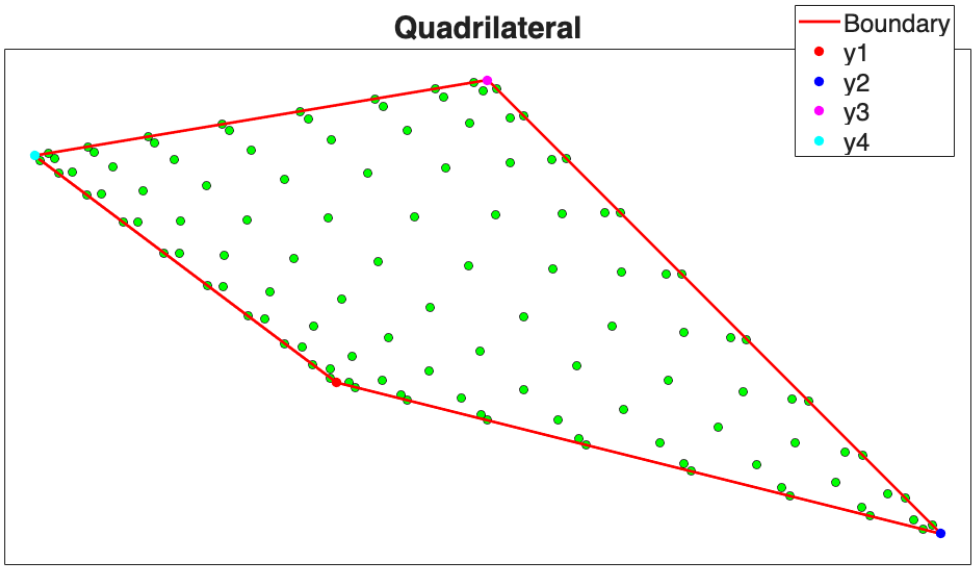

```
InfSpace
  The whole space in Cartesian coordinates.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   L1 - a mapping parameter such that half of the y1 points lie in
   [-L1 + y10, L1 + y10]
   L2 - a mapping parameter such that half of the y2 points lie in
   [-L2 + y20, L2 + y20]
   (y10 = 0) - the 'origin' in y1
   (y20 = 0) - the 'origin' in y2
```



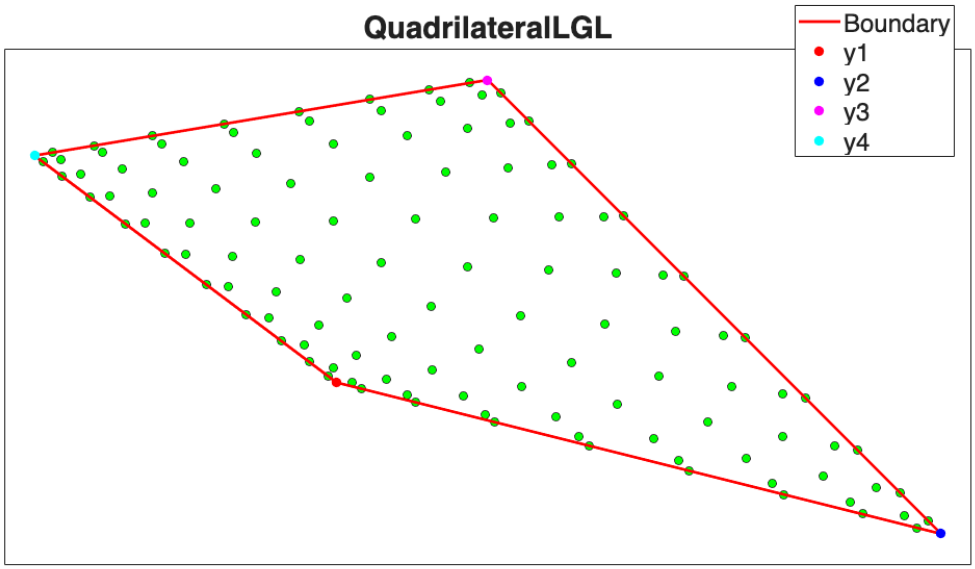

```
Triangle
  A triangle specified by the three corners, y1, y2, y3.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   Y = [y1;y2;y3] - with yi being 1x2 vectors of the Cartesian
   coordinates of the corners
```



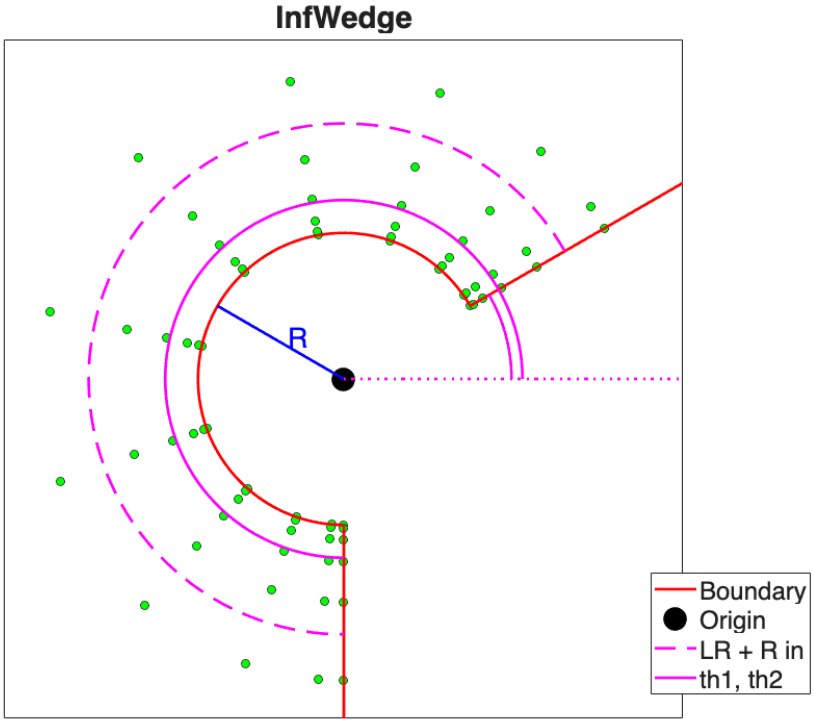

```
Quadrilateral
  A _convex_ quadrilateral specified by four corners, y1, y2, y3, y4.

  Inputs:
   N - [N1,N2] the number of points in the two coordinates
   Y = [y1;y2;y3;y4] - with yi being 1x2 vectors of the Cartesian
   coordinates of the corners
```



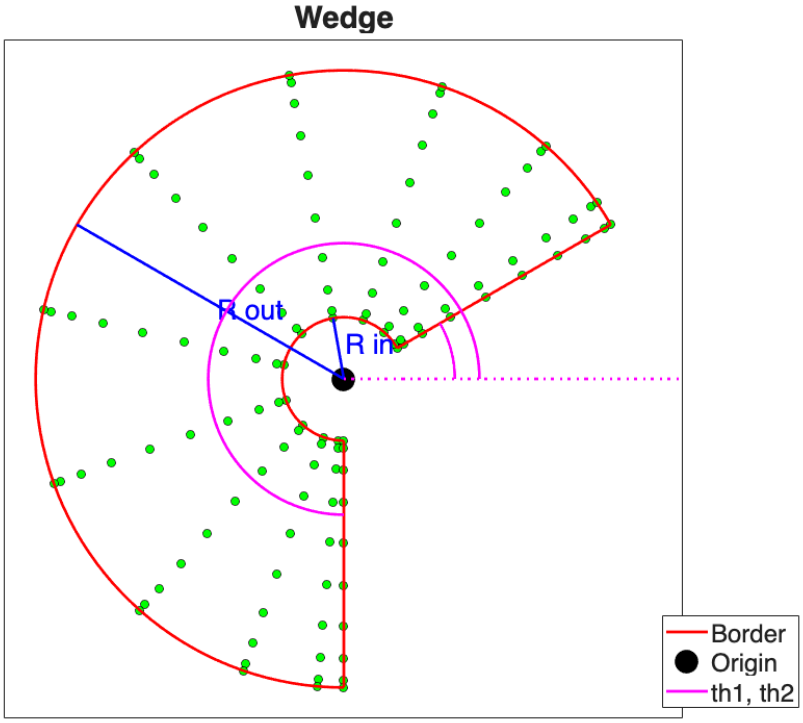
QuadrilateralLGL
A _convex_ quadrilateral specified by four corners, y1, y2, y3, y4.
Note: this is exactly the same as QUADRILATERAL except we use
Legende-Gauss-Lobatto nodes and weights. If in doubt, use
QUADRILATERAL.

Inputs:
N - [N1,N2] the number of points in the two coordinates
Y = [y1;y2;y3;y4] - with yi being 1x2 vectors of the Cartesian
coordinates of the corners




```
InfWedge
  An infinite wedge of a disc, between th1 and th2, centred at Origin.
  Can also be used to create a wedge of an annulus by specifying R_in,
  the inner radius.

  Inputs:
    N - [N1,N2] number of points in the two coordinates
    LR - mapping parameter for radial coordinate; half the points are
         in [R_in,R_in+LR]
    th1, th2 - the start and end angles
    (Origin = [0,0]) - origin of the circle
```




```
Wedge
  A wedge of a disc of radius R, between th1 and th2, centred at
  Origin.
  Can also be used to create a wedge of an annulus by specifying R_in,
  the inner radius.

  Inputs:
    N - [N1,N2] number of points in the two coordinates
    R (or R_out) - (outer) radius of the disc
    th1, th2 - the start and end angles
    (Origin = [0,0]) - origin of the circle

  Alternatively, specify:
    y1Min - R_in
    y1Max - R_out
    y2Min - th1
    y2Max - th2
```

Published with MATLAB® R2025b